

Operational Analytics & Metric Spike Investigation

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1. Project Overview

This project presents a dual case study using SQL to investigate two distinct business scenarios: operational efficiency and user behavior patterns. The primary goal was to extract actionable insights from raw data and present them in a clear, accessible format. The project initially concluded with a static report and was later extended to include dynamic, interactive dashboards to provide a more powerful analytical tool.

The project is divided into two parts:

- **Case Study 1: Job Data Analysis:** Focused on analyzing operational performance, reviewer throughput, and data quality.
- **Case Study 2: Metric Spike Investigation:** Aimed at understanding user behavior, engagement trends, and the drivers behind significant metric spikes.

2. Tech Stack

The following technologies were used to accomplish this end-to-end analysis:

- **MySQL Workbench:** Used for database creation, importing CSV datasets, and performing all SQL-based analysis.
- **Microsoft Power BI:** Used for data modeling, creating DAX measures, and developing two interactive dashboards for dynamic data exploration.
- **Microsoft Excel:** Initially used for creating charts from SQL query outputs.

3. Analytical Approach

A systematic approach was followed to move from raw data to final insights:

1. **Problem Understanding:** The business requirements for analyzing job data and user metrics were clearly defined.
2. **Data Exploration:** The structure of the source tables and the relationships between them were studied to inform the querying strategy.
3. **SQL Querying:** Complex queries involving CTEs, window functions, and aggregations were written to extract key metrics.
4. **Insight Generation:** The analysis focused on key performance indicators (KPIs) such as throughput, retention, and user engagement.
5. **Interactive Dashboard Development:** The aggregated data and analyses were visualized in Power BI to create two purpose-built, interactive dashboards that allow for dynamic filtering and exploration.

4. Case Study 1: Job Data Analysis

Objective: To analyze the performance and throughput of job reviewers to identify efficiency trends and data quality issues.

Key SQL Analyses & Insights:

- **Jobs Reviewed Over Time:** Analysis revealed that reviewer activity peaked on November 28, 2020.
- **Throughput Analysis:** Using a 7-day rolling average proved more effective for understanding long-term performance trends by smoothing out daily volatility.
- **Language Share:** Persian was the most reviewed language, accounting for 37.5% of the jobs, providing a clear basis for resource allocation.
- **Data Quality:** The detection of duplicate rows highlighted a need for improved data validation processes to ensure analytical accuracy.

Interactive Dashboard: Operational Performance

An interactive Power BI dashboard was built to monitor these metrics dynamically.

- **Key KPIs:** The dashboard prominently features summary cards for **Weekly Average Throughput** and **Max Job Reviewed Per Hour**.
- **Core Visuals:** It includes a bar chart for **Daily Average Throughput**, a horizontal bar chart ranking **Jobs Reviewed Per Hour** by date, and a **Language Share** donut chart.
- **Interactivity:** A date slicer allows managers to filter the entire report for a specific time frame to analyze performance during that period.

5. Case Study 2: Investigating Metric Spikes

Objective: To analyze user activity data to understand engagement patterns, growth trends, and the reasons behind a significant spike in metrics.

Key SQL Analyses & Insights:

- **Weekly User Engagement:** A major spike in user activity was identified around July 27, 2014, suggesting a successful marketing campaign or product update.
- **User Growth:** The analysis showed a strong upward trend in user acquisition, with August 2014 being the highest month for growth with 1031 new users.
- **Weekly Retention:** Retention analysis indicated a significant drop-off after the first week, suggesting that the user onboarding experience could be improved.
- **Device & Email Engagement:** The **MacBook Pro** was the most used device among engaged users. The `sent_weekly_digest` was the most frequent email action, indicating its importance in driving user interaction.

Interactive Dashboard: Metric Spike Investigation

A comprehensive Power BI dashboard was developed to explore user behavior interactively.

- **Key KPIs:** High-level metrics are displayed at the top, including **Total Events**, **Total Actions**, **Average Retained Users**, and **Average of New Users**.
- **Core Visuals:** The dashboard features trend charts for **Event Count by Date** and **New Users by Month**, along with breakdowns of **Events Count by Device** (pie chart) and **Device Actions by Count** (bar chart).

- **Interactivity:** A date range slider enables product and marketing managers to isolate specific time periods—such as the week of a metric spike—to see how all other metrics were affected.

6. Project Impact and Conclusion

This project successfully provided data-driven insights that can inform strategic decisions. The operational analysis offers a clear path to improving resource allocation and data integrity. The user behavior analysis helps in targeting high-engagement segments (like MacBook Pro users) and optimizing marketing strategies to enhance user retention.

By extending the initial SQL analysis into interactive Power BI dashboards, the project's value was significantly enhanced, transforming static findings into dynamic, exploratory tools for ongoing business monitoring.